PARK ASSIST SYSTEM [WITH CENTER DISPLAY]

id0920zz011000

Caution

- **The park assist system is a supplementary system which displays the conditions at the rear of the vehicle when reversing. Always directly confirm the safety of the surrounding area visually when reversing.**

Outline

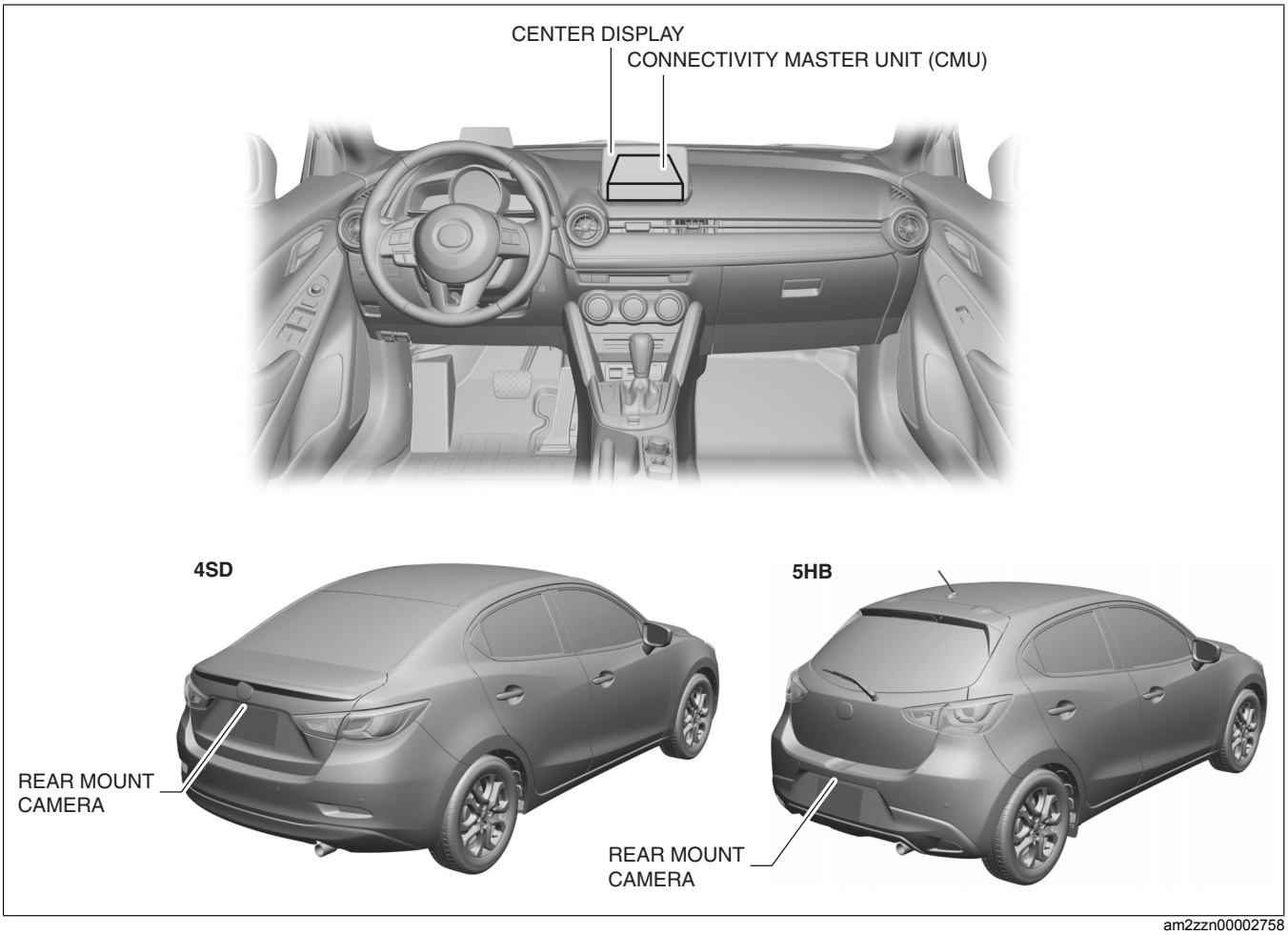
- When the vehicle is reversing, the rear mount camera installed to the trunk lid (4SD)/rear bumper (5HB) displays the images at the rear of the vehicle in the center display for verification of the presence of pedestrians and obstructions at the rear of the vehicle.

Function

Guide line display

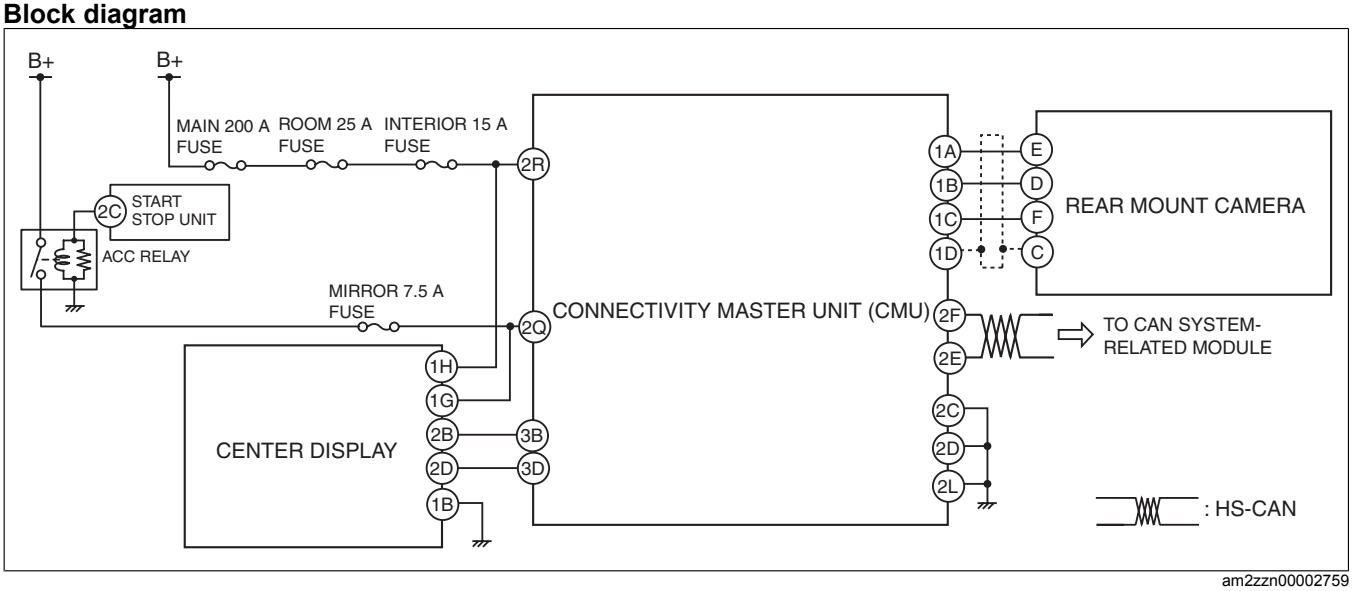
- The rear mount camera displays the distance reference lines and vehicle width extension lines based on the images shot at the rear of the vehicle by the camera.

Structural View



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Block diagram

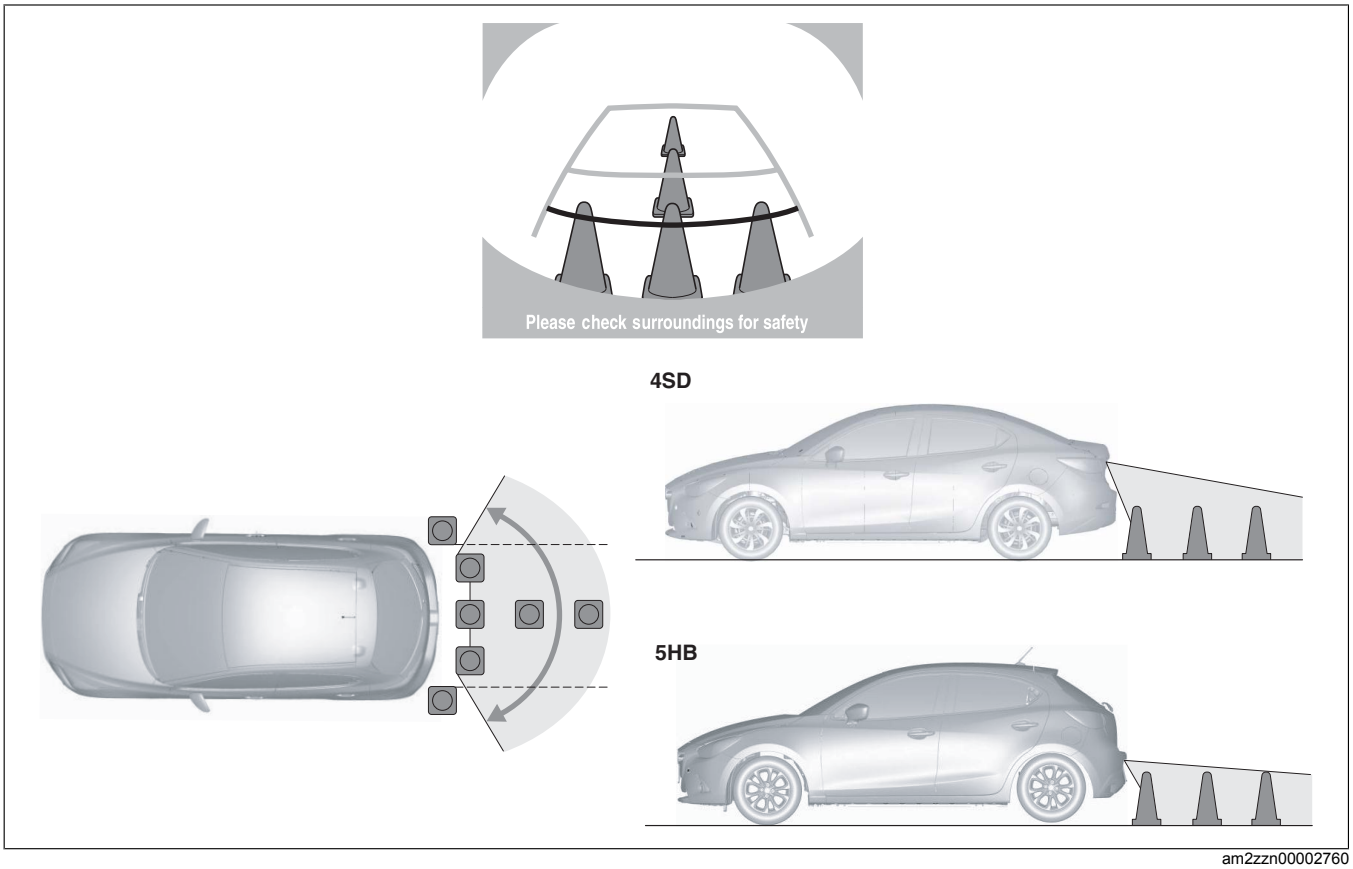


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Image range

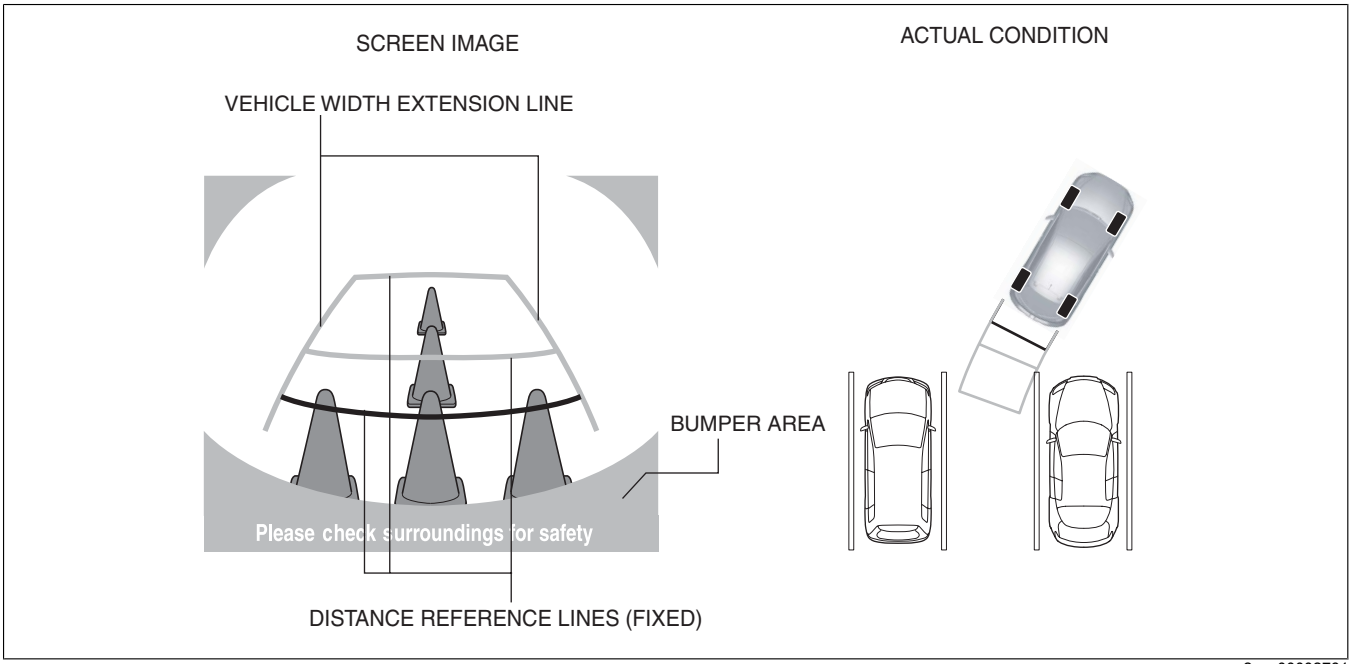
Note

- The image range may differ depending on the road surface conditions and the vehicle conditions. In addition, images near both sides of the rear bumper and below the rear bumper cannot be obtained.



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Rear view monitor screen



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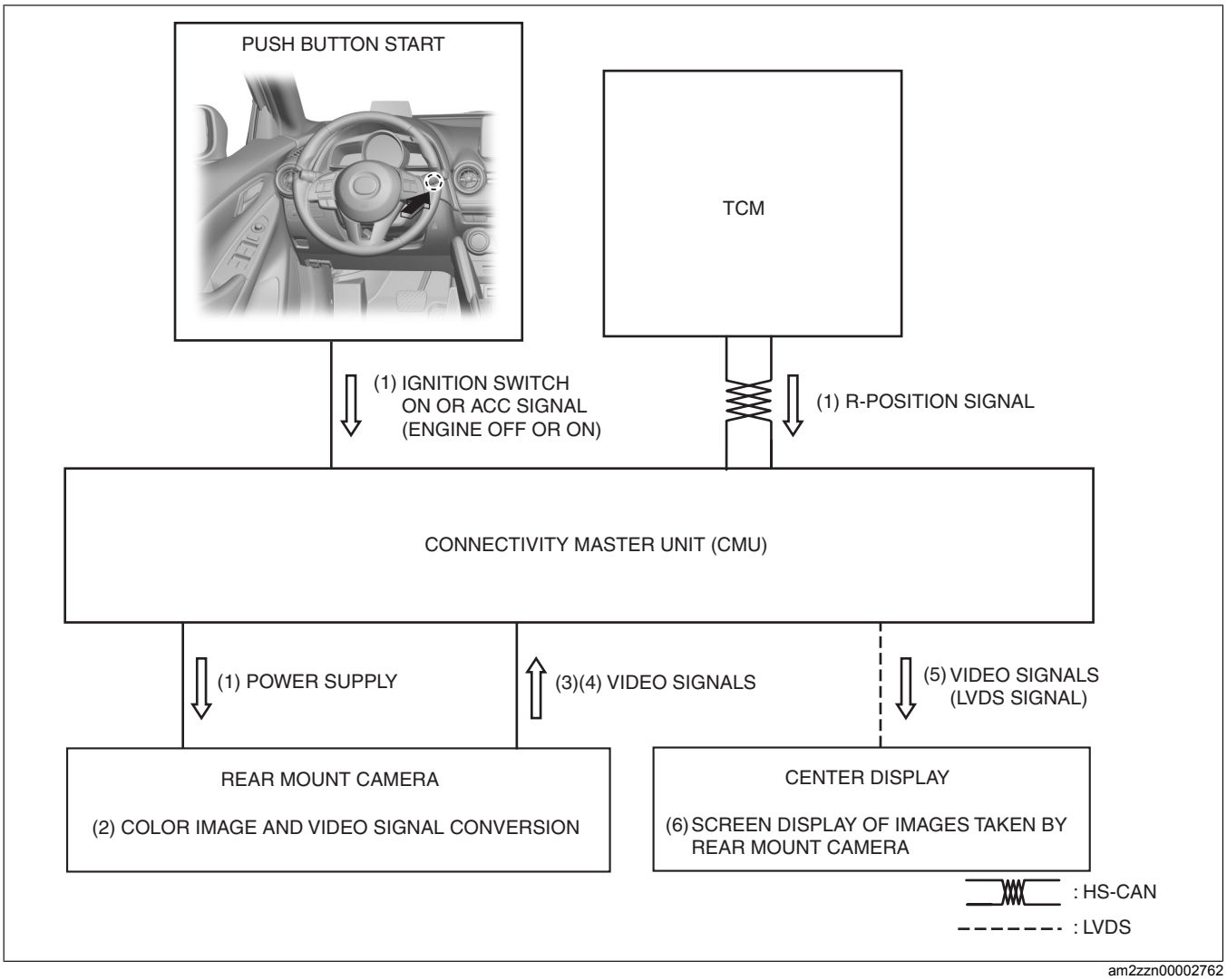
Display item	Display line color	Function
Distance reference lines (fixed)	Red	• Shows distance reference line **0.5 m** from the rear of the rear bumper. • Not in conjunction with steering operation.
	Yellow	• Shows distance reference line **1.0 m** from the rear of the rear bumper. • Not in conjunction with steering operation.
	Yellow	• Shows distance reference line **2.7 m** from the rear of the rear bumper. • Not in conjunction with steering operation.
Vehicle width extension line	Yellow	• Vehicle width extension line.
Bumper area	Black	• Displays the rear bumper area. **Note** **5HB** • The pseudo rear bumper area is masked in black because the rear end of the rear bumper cannot be picked up from the rear mount camera position.

Operation

Caution

- **Images displayed by the rear mount monitor camera differ depending on the actual conditions resulting from road surface conditions, the number of passengers, and cargo load. Even though it may appear that the vehicle will not come into contact with an object as seen in the image, it may actually come into contact with it. When reversing the vehicle, always verify the surroundings visually.**

1. When the ignition is switched to ACC or ON (engine off or on) and an R position signal from the selector lever is received (1), the CMU supplies (1) power to the rear mount camera.
2. The rear mount camera shoots the conditions at the rear of the vehicle in color which is converted (2) to a video signal.
3. The rear mount camera combines the distance reference lines (fixed lines) and the vehicle width extension lines with the camera images and converts (3) them to a video signal.
4. The rear mount camera sends (4) the converted video signal to the connectivity master unit (CMU).
5. When a video signal is received from the rear mount camera, the CMU sends the received (5) video signal (LVDS signal) to the center display.
6. The center display displays (6) the images at the rear of the vehicle taken by the rear mount camera based on the received video signal (LVDS signal).



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Note

- It may be difficult to see the display under the following conditions; however, it does not indicate a malfunction:
  - — When the temperature around the lens is high/low.
  - — When the camera is wet such as on a rainy day or during period of high humidity.
  - — When foreign material such as mud is stuck around the camera.
  - — When the camera lens reflects sunlight or headlight beams.